# ABDUL ALEEM SHAIK GADDA

PhD Student (1st Year)

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## PROFESSIONAL PROFILE

Highly motivated and dedicated biomedical engineering researcher with a strong background in programming, mathematics, and engineering. Enthusiastic about applications of Artificial Intelligence in health care to improve people's lives. Always ready to learn new technologies and design practical experiments to integrate AI in daily life for monitoring, diagnostic and treatment purposes.

## SKILLS

**Technical:** Python, MATLAB, LaTeX, Julia.

**Languages:** English, Telugu, Hindi

**Toolboxes:** Freesurfer, MIPAV, Brain Suite

**Equipment:** Confocal Microscope (Zeiss)

## RESEARCH INTERESTS

- Artificial Intelligence, Brain-Computer Interfaces, Signal Processing and Analysis, Neural Robotics, Computational Neuroscience, Cognitive Neuroscience, Biomedical Data Science, Rehabilitation, Assistive technology

## EDUCATION

2026 - 2031 **Rochester Institute of Technology** **CGPA: -.–/4.00**
Computing and Information Sciences, PhD

2020 - 2025 **Indian Institute of Technology (BHU) Varanasi** **CGPA: 9.65/10.00**
Biomedical Engineering, IDD (B.Tech in Bioengineering + M.Tech in Biomedical Technology)

## WORK EXPERIENCE

Dec 2025 **Realtime Word prediction through EMG signals** **Online, IIT Kanpur.**

- Active project guide to build real time deployable machine learning models to predict words using EMG signals acquired.
- Developed hybrid deep learning and statistical ML models that delivered stable probabilistic predictions while reducing bias through calibration and regularization techniques.

Jun '25 - Jan '26 **Interplay of Sensory and Place cells in social memory development** **Max Planck Institute, Germany.**

- Research Scientist at the RoLi Lab, Max Planck Institute for Biological Cybernetics, Tübingen, Germany.
- Experimentations and analysis on the social cognitive behaviour of larval zebrafish for spatial specific place cells based on memory development in the presence of conspecific species.
- Main objective was to identify and distiguish between the activity of place cells and sensory cells in response to the presence of similar fish within a fixed boundary space.

May - Jul 2024 **FidCC.AI - Fidelity Covered Call through AI** **Fidelity Investments, India.**

- Research Intern at Artificial Intelligence Center Of Excellence, Fidelity Management and Research India.
- Developed a Covered Call Option investment strategy simulation product using LSTM and Black Scholes Model, and streamlit.
- Main objective was to help the customers to use AI genertaed insights and predictions for financial strategy deployments.

May - Aug 2023 **System Identification of Spinal Cord to Quantify Muscle Response Dynamics.** **University of Alberta, Canada.**

- MITACS Globalink Research Intern 2023.
- Employed the system identification techniques in MATLAB, to identify the transfer function of the spinal cord system that is responsible to convert the electrical stimulation impulse provided to the spinal coed into a muscle response.
- Optimum performance of the spinal cord model was observed for a 7th order system with an overlap of more than 85%.

Jan - Sep 2022 **Understanding cognitive processing under degraded viewing conditions** **KU Leuven, Belgium**

- Research Internship
- Worked under supervision of Dr. Marc Van Hulle (KU Leuven), Arno Libert (KU Leuven), Pergher Valentina (Harvard University) on Cognition and Memory co-relation with frequency bands, and their activity according to different stimuli, difficulty levels of the cognitive task "N - BACK".

## PROJECTS

2024 - 2025 **Optimal Preictal Interval Identification For Epileptic Seizure Prediction using ML** **Inidian Institute of Technology (BHU) Varanasi**

- ***Masters Thesis Project Supervisor : Dr. A R Jac Fredo***
- Pre-processed the TUH, Siena datasets using the developed pipeline in the previous studies and deployed state of the art Machine Learning algorithms to classify the preictal state from the interictal state of an occuring epileptic seizure.
- Identified the Optimal preictal interval across various feature domains and combinations using multiple statistical ML algorithms.

Jan - May 2024 **Epileptic Seizure Prediction using Wavelet features and Machine Learning** **Inidian Institute of Technology (BHU) Varanasi**

- ***Supervisor : Dr. A R Jac Fredo***
- Pre-processed the TUH Dataset using the developed pipeline in the previous studies and deployed state of the art Machine Learning algorithms to classify the preictal state from the interictal state of an occuring epileptic seizure.

Oct - Dec 2023 **Automated Emotion Recognition Using Deep Learning** **Inidian Institute of Technology (BHU) Varanasi**

- Deployed different Deep Learning Architectures to classify the emotions (Base, Fun, Stress) from the WE-SAD dataset.
- Time - Frequency respresentations were obtained using STFT, CWT, MFC decomposition methods and were fed as inputs to different architectures such as cCNN, VGG16, AlexNet.
- Highest classification accuracy observed was 85% using the VGG16 architecture.

Aug - Nov 2023 **Analysis of Functional connectivity in different Brain regions** **Inidian Institute of Technology (BHU) Varanasi**

- Led a group of three undergraduate students.
- Determined the best correlation technique among Mutual Information, R squared, Pearson Correlation to examine connections in various brain regions and categorise individuals with ASD and TD.
- Individuals were classified with an accuracy of over 80%.

Jan - May 2023 **Human Machine Interface Using EMG Signals** **Inidian Institute of Technology (BHU) Varanasi**

- The objective was to generate mouse click using gestures classified using Electromyograph signals collected using Arduino UNO Microcontroller.
- Classification of the EMG signals was done using XGB classifier in realtime by interfacing the microcontroller with python.

Dec '21 - Jan '23 **Optimization of Preictal time Interval for Seizure Prediction** **Inidian Institute of Technology (BHU) Varanasi**

- Designed a research pipeline, processed the electroencephalogram signals obtained from the publicly available CHB-MIT dataset and classified the preictal state of signal with the interictal state to predict the onset of a seizure using machine learning algorithms.
- A prediction window horizon of 10 minutes was observed.

Jul - Nov 2021 **Volumetric Segmentation of Sub-Corticular Brain regions** **Inidian Institute of Technology (BHU) Varanasi**

- Region of Interest segmentation using 3D volumes, by reconstructing the raw MRI data for the volumetric analysis in Freesurfer. And, manually segmenting volumes of interest (Corpus Callosum, Brain stem) using MIPAV.

## PERSONAL PROJECTS

Sept - Oct 2024 **Amazon Machine Learning Hackathon** **Kaggle Competition**

- Fine-tuned a Visual QnA Large Language Model to extract the dimensional related information from the product image based on the user input query.
- The dataset consisted of product images with unlabelled product dimensions, making the challenge a perfect application on fine-tuning Visual QnA architectures.

Oct - Nov 2024 **IEEE SMC BR4IN.IO 2024 Hackathon** **IEEE SMC 2024 Hackathon**

- The objective was to classify the hand gestures (Rock, Paper, Scissors, Relaxed) using the electrocorticogram (ECoG) signals recorded in 30 trials.
- Dimensionality reduction was achieved by using the PSD in the 50 - 300 Hz High Gamma frequency region, selecting top 10 highly active channels and passing them into the 1D CNN configurable classifier.
- 4 Class classification accuracy achieved was 80% including the relaxed state, and 100% for the 3 class gesture classification using a 5 fold Cross-Validation approach.

Ongoing **Fine-tuning Generative AI for Medical data Analysis** **Inidian Institute of Technology (BHU) Varanasi**

- Fine-tuning Open source LLMs to prcess Electronic Health Records to provide contextual based responses for the prospective user.
- Deploying pre-trained Vision Transformers to identify and lock the spatial importance of medical images for the Visual QnA architectures.

## WORKSHOPS

- Participated in SPARC sponsered International Workshop on Neurobiology of Pain & Itch.
- Participated and contributed to SERB sponsored one-day training program on Introduction to Computational Neuroscience.

## PARTICIPATIONS AND ACHIEVEMENTS

- Acquired **2nd Winner** position in the IEEE SMC 2024 Brain-computer interface hackathon, focusing on the classification of rock, paper, scissor gestures with an unique approached driving towards the real time applications for faster calibrations.
- Participated and acquired **2nd Position** in "AL-VTOLA", an event in techfest conducted by Indian Institute of Technology Bombay (IITB), designed an unmanned aerial vehicle for efficient surveillance over an area of 50sq.km.
- One of the **top 10 finalists** in the Hult Prize Event On-Campus Finals, member of team Merci, presented an idea focused on developing a device that increases the food preserving capacity of home refrigirators.
- **Finalist** in the **Abhipraya'20**, a case study event conducted by the Social Service Council of Indian Institute of Technology (BHU) Varanasi, **leader of team ENYLEC**, in which the motive was to pitch the idea of an efficient and renewable energy automobile.

## JOURNAL PUBLICATIONS

- **A. Aleem Shaik Gadda**, A. Chahar, Y. Rajamanickam, J. Thomas, and J. Fredo Agastinose Ronickom, "Data-Driven Quantification of the Pre-Seizure Period in Epilepsy Using Scalp EEG: A Causal Observational Study" (submitted to) IEEE Transactions on Neural Systems and Rehabilitation Engineering.
- P. Sriram Kumar, P. K. Govarthan, **A. Aleem Shaik Gadda**, N. Ganapathy and J. Fredo Agastinose Ronickom, "Deep Learning-Based Automated Emotion Recognition Using Multimodal Physiological Signals and Time-Frequency Methods," in IEEE Transactions on Instrumentation and Measurement, vol. 73, pp. 1-12, 2024, Art no. 2526912, doi: 10.1109/TIM.2024.3420349.
- A. Bhattacharya, G. Manoj, V. Gupta, **A. Aleem Shaik Gadda**, D. Vedantham, A. P. A. Prince, P. Rani, A. Ramaniharan, and J. Fredo, "Comparative evaluation of geometrical, Zernike moments, and volumetric features of the corpus callosum for discrimination of ASD using machine learning algorithms," Int. J. Biomed. Eng. Technol., vol. 43, pp. 275-296, 2023, doi: 10.1504/IJBET.2023.134588.
- G. Manoj, V. Gupta, A. Bhattacharya, **A. Aleem Shaik Gadda**, D. Vedantham, and J. F. Agastinose Ronickom, "Diagnostic classification of autism spectrum disorder using sMRI improves with the morphological distance-related features compared to morphological features," Multimedia Tools Appl., pp. 1-22, Mar. 2024, doi: 10.21203/rs.3.rs-2277683/v1.

## CONFERENCE PUBLICATIONS

- **A. Aleem Shaik Gadda**, D. Vedantham, J. Thomas, Y. Rajamanickam, R. N. Menon, and J. Fredo Agastinose Ronickom, "Optimization of Preictal Interval Time Period for Epileptic Seizure Prediction Using Temporal and Frequency Features," Stud. Health Technol. Inform., vol. 302, pp. 232-236, May 2023, doi: 10.3233/SHTI230109.
- Aditi Bhattacharya, Gokul Manoj, Vaibhavi Gupta, **A. Aleem Shaik Gadda**, Amalin Prince A, Priya Rani, Jac Fredo AR 'Evaluation of Zernike moments of corpus callosum for discrimination of autism using Random Forest', XXII International Conference on Mechanics in Medicine and Biology, Bologna, Italy, September 19-21, 2022. (Abstract Publication)

## CONFERENCE PRESENTATIONS

- **Abdul Aleem Shaik Gadda**, Abirami S, John Thomas, Yuvaraj Rajamanickam and Jac Fredo Agastinose Ronickom, "Epileptic Seizure Prediction using Wavelet Features and Machine Learning,"XXIII International Conference on Mechanics in Medicine and Biology – ICMMB 2024, Bruxelles, Belgium, September 11-13, 2024. **Presenter: Dr. A R Jac Fredo**

- Reshma K, Smarth S, **Abdul Aleem Shaik Gadda**, Pragya, Sabitha R, Jac Fredo AR and Ramakrishnan S, 'Identification of self-limiting samples from the high-risk cervical cancer using microarray data and unsupervised machine learning algorithms', Emerging Trends in Translational Bioinformatics Conference, BITS Mesra, Ranchi, India, December 5-7, 2024. **Presenter: Reshma K**

- Pragya, **Abdul Aleem Shaik Gadda**, and Jac Fredo Agastinose Ronickom, 'Drug absorption property prediction using molecular descriptors and machine-learning algorithms for drug repurposing in PDAC,' Emerging Trends in Translational Bioinformatics Conference, BITS Mesra, Ranchi, December 5-7, 2024. **Presenter: Pragya**

## DECLARATION

I, Abdul Aleem Shaik Gadda, hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

Date : 23 Sep 2026
Place: Rochester